

Global Equity Risk Premiums (ERP) & Country Risk Premiums (CRP)

Methodology Paper

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Abstract. Every valuation rests on a discount rate, and every discount rate rests on a risk premium. When that valuation involves a company operating in Brazil or an acquirer pricing a target in Indonesia, the question of which risk premium to use becomes anything but trivial, investors in these markets bear political instability, violence, corruption, and institutional fragility that mature markets have largely priced away. Ignoring that is not conservatism, it is mispricing. This paper documents the methodology for estimating Equity Risk Premiums and Country Risk Premiums across 174 countries and 9 global regions as of April–May 2026, building explicitly on the framework developed by Professor Aswath Damodaran of NYU Stern, with three methodological refinements: the risk-free rate is derived by netting CDS spreads against Switzerland rather than applying a rating-based default spread, ensuring consistency with the CRP framework; the mature market ERP strips America's full country risk premium rather than its raw sovereign default spread from the implied ERP; the relative volatility multiplier uses indices with balanced exposure across both developed and emerging markets; all through a three-tier hierarchy of CDS spreads, Moody's rating-implied spreads, and Political Risk Service scores. The result is a set of estimates any analyst can use in cross-border valuation without rebuilding the methodology from scratch.

1. Introduction

The foundation of modern finance rests on a deceptively simple observation, investors are risk averse. Given two investments with identical expected returns, a rational investor will always prefer the less risky one. For riskier investments to attract capital, they must offer a higher expected return, a premium above what an investor could earn

by simply parking money in a risk-free asset. This excess return is what we call the Equity Risk Premium (ERP).

The ERP sits at the heart of virtually every valuation and capital allocation decision. It drives the hurdle rate in corporate finance, anchors the cost of equity in a discounted cash flow model, and sets the benchmark against which portfolio managers assess whether risk is being rewarded. Despite its centrality, the ERP is not directly observable, it must be estimated, and the method of estimation matters enormously.

The most intuitive approach is the historical one, look back at what equities have actually returned above the risk-free rate, and use that as your estimate going forward. For economies with long capital market histories, it has real appeal. But the method carries burdens that compound quickly. How far back should one look? Each window captures a different economic regime, a different rate environment, a different global order. Consider the Nifty 50's own return history:

Table 1: Nifty 50 Historical Returns

Period	arithmetic mean	geometric mean
5 Years	20.52%	17.72%
10 Years	11.75%	9.15%
25 Years	14.35%	10.62%

The 5-year window produces an arithmetic mean of 20.52%, extend to 25 years and it falls to 14.35%. Which number is the "true" equity risk premium? There is no principled answer, each is equally valid and each lead to a materially different cost of equity and, ultimately, a different valuation. The choice of window alone can swing an estimate by hundreds of basis points.

The averaging method compounds this further. The arithmetic mean overstates the compounded return an investor actually earns over time, the geometric mean understates the expected one-period premium. As the table above illustrates, the gap between the two can exceed three percentage points, enough to render a project acceptable under one estimate and unattractive under another. And both are backward-looking by construction: they tell you what risk was rewarded in the past, not what investors are demanding today.

These limitations motivate the forward-looking, implied approach, estimating ERP the same way one estimates the yield-to-maturity on a bond. Take the current market price, project expected cash flows, and solve for the discount rate that reconciles the two.

Applied to a broad equity index like the S&P 500, this produces a dynamic, real-time estimate of what the market is collectively pricing as compensation for equity risk, with no assumption required about which historical window is representative. When this method is applied to a mature and developed market, one with deep capital markets, reliable earnings histories, and minimal sovereign risk, the resulting estimate is what we call the mature market ERP. It represents the baseline risk premium for equity investment in the safest, most transparent markets in the world, and it serves as the

anchor for all further risk premium estimation. Every other market in the world is, in some sense, priced relative to this baseline.

The natural next question is: why use it as a baseline? why not simply estimate the implied ERP for the country where the company is actually operating?

The honest answer is, for a handful of countries, you can. Markets with sufficient depth, liquidity, and analyst coverage can support a credible implied ERP estimate of their own. But those markets are the exception, not the rule. The vast majority of the world's equity markets fail on one or more of the preconditions that make the implied method reliable in the first place.

Consider what the calculation actually requires: a broad, liquid index with reliable earnings forecasts from which to project future cash flows, a stable payout history from which to anchor near-term distributions.

Frontier and emerging markets routinely violate all of these conditions simultaneously. As price discovery is thin and episodic, often dominated by a handful of large-cap names. Analyst coverage thins dramatically below the top decile of listed companies. Earnings forecasts, where they exist at all, are less anchored in the kind of institutional consensus that makes S&P 500 projections relatively stable. And the local risk-free rate is not risk-free at all. It already embeds a sovereign default premium, if not stripped any implied ERP computed against it double-counts the very risk we are trying to isolate.

There is a further, subtler problem specific to high-growth emerging markets. Even where the implied method is mechanically executable, where prices are observable and forecasts exist, the resulting estimate tends to be misleadingly low. The implied ERP is solved by finding the discount rate that reconciles current market prices with projected cash flows. In markets like India, where consensus growth forecasts are often anchored more to narrative than to fundamentals, it is the growth assumption that does the heavy lifting in justifying elevated prices.

So, why not simply apply the same implied ERP to every country and be done with it?

Can't do that. Because as mature-market ERP is insufficient for the task at hand. An analyst valuing a company in Indonesia, a fund pricing an equity stake in Nigeria, or an acquirer building a DCF for a target in Brazil faces risks that no US-calibrated risk premium can capture. As sovereigns rated below 'Aaa' is expected to carry additional multi-dimensional country risks for which mature market implied premium alone cannot account for, as these risks manifests through political structure: where changes in policies cannot be taken for granted, exposure to violence: increases cost by driving up operating and insurance costs, corruption: which functions as an implicit tax that quietly erodes cash flows and value, and the quality of legal systems: where weak enforcement property rights undermines the very foundations of a business.

This is the problem the Country Risk Premium (CRP) addresses. The CRP is the incremental return investors demand above the mature-market baseline to compensate

for the specific risks of investing in a given country. And it is a practical necessity for anyone doing cross-border valuation work. Formally:

$$ERP(\text{country}) = ERP(\text{mature market}) + CRP(\text{country})$$

This paper documents the methodology for estimating both the ERP and CRP across 174 countries in 9 global regions, building on the framework developed by Professor Aswath Damodaran of NYU Stern. The goal is not to produce a black box but is to make the assumptions explicit, the hierarchy transparent, and the output usable by anyone doing cross-border valuation work without having to rebuild the methodology.

2. Departures from the Damodaran Framework

This study builds on the framework developed by Professor Aswath Damodaran of NYU Stern and introduces three methodological refinements.

a. Risk-Free Rate Construction

Damodaran adjusts the US Treasury yield using a rating-based default spread for the US sovereign of 26 bps. This study uses the US CDS spread net of Switzerland of 40 bps instead. The difference is not material in basis points but matters for internal consistency, if CDS spreads are the primary signal throughout the CRP framework, applying a rating-based adjustment for the risk-free rate creates an inconsistency at the model's foundation. This study eliminates it.

Table 2: Implications of Changes in Default-Adjusted Risk-Free Rate

	Default Spread	Risk-Free Rate	Implied ERP
This Study	0.40%	4.21%	4.00%
Damodaran	0.26%	4.35%	3.99%

b. Mature Market ERP

Damodaran derives the mature market baseline by subtracting the raw US default spread from the implied ERP. This study subtracts the full US country risk premium, the default spread scaled by the relative volatility multiplier on the basis that equity investors bear equity-scaled country risk, not bond-market default risk. The adjustment produces a mature market ERP of 3.00% using 1% CRP, versus using 0.26%, which would result in 3.74%.

Table 3: Implications of Changes in Spread for Mature Market ERP

	Additional Volatility Spread	Mature Market ERP	World ERP
This Study	1.00%	3.00%	5.99%
Damodaran	0.26%	3.74%	6.73%

c. Relative Volatility Multiplier

Damodaran estimates the multiplier using the S&P Emerging BMI Index against the iShares JP Morgan USD Emerging Markets Bond ETF, both emerging-market instruments together yielding a multiplier of 1.53x. This study uses the iShares MSCI ACWI ETF (23 developed, 24 emerging markets) against the JPMorgan Global Aggregate Bond Fund, reflecting the perspective of a globally diversified investor rather than an emerging-market-only one. This yields a multiplier of 3.03x.

Table 4: Implications of Changes in Relative Volatility

	Relative Volatility	World CRP	World ERP
This Study	3.03x	2.99%	5.99%
Damodaran	1.53x	1.52%	5.01%

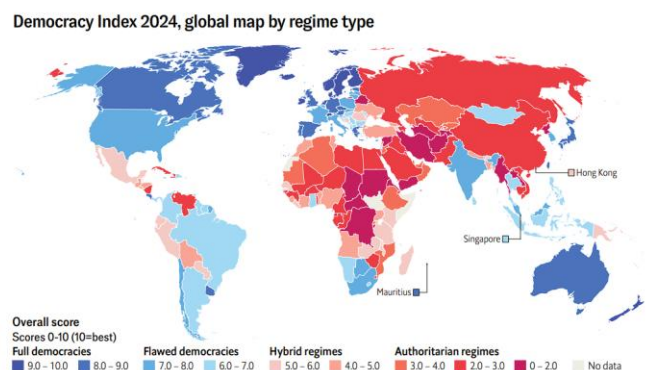
3. Country Risks

Country risk is not a single number, but a composite of several distinct forces, each of which affects businesses in different ways. Professor Damodaran identifies four dimensions that matter most in practice

a. Political Structure

The relationship between political systems and business risk is less straightforward than it first appears. The instinct is to say democracies are safer, and for the most part that holds, but the risk profile is different, not absent. In a democracy, newly elected governments can reverse, modify, or discard policies their predecessors put in place. Businesses have to continuously adapt. In an authoritarian regime, policy continuity can be surprisingly stable. When change comes in an authoritarian system, it tends to be abrupt and wrenching rather than gradual. The risk is discontinuous rather than continuous. Neither system eliminates political risk; they just deliver it differently.

Figure 1: World Democracy Index 2025

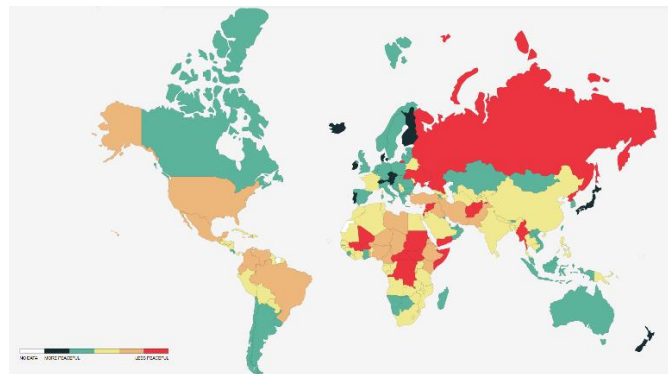


Source: Economist Intelligence Unit

b. Exposure to Violence

Violence affects businesses through multiple channels simultaneously. There is the direct operational threat, but also the costs that accumulate around it: security for employees and facilities, insurance premiums where coverage is even available, and the higher return that any rational investor demands for the uncertainty. The Middle East, parts of Central Africa, and the states surrounding the Russia-Ukraine conflict remain the highest-risk regions on this dimension.

Figure 2: Global Peace Index 2025

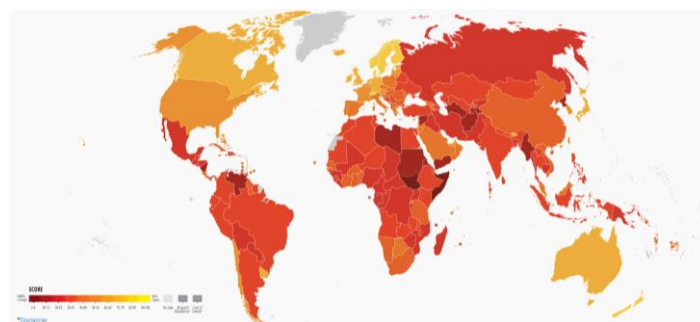


Source: Institute for Economics & Peace

c. Corruption

For businesses that refuse to engage with corruption, it creates a structural disadvantage against competitors who will. For those that do engage with it, it introduces implicit costs in form of tax, legal exposure, and unpredictability that erode value quietly over time. It is most entrenched in environments where businesses require licenses and approvals for routine activities, giving the officials who control those approvals leverage over the businesses they regulate. As the map makes clear, this is not a problem confined to a handful of outlier countries.

Figure 3: Corruption Perceptions Index 2025



Source: Corruption Perceptions Index 2025

d. Legal Systems

The quality of a legal system is not just about what laws exist on paper; it is about how reliably and quickly they get enforced. A legal system where contract disputes take a decade to resolve is, in practical terms, barely better than one that is openly biased. The ability to enforce ownership rights, hold counterparties to agreements, and seek recourse when things go wrong is foundational to running any business. Where that foundation is weak, the risk premium required to invest there should reflect it.

4. Estimating the S&P 500 Implied ERP

The implied ERP for the S&P 500 is estimated using a dividend discount model applied to the index level, the same logic as solving for the yield-to-maturity on a bond, except the cash flows are equity distributions rather than coupons, and the "price" is the current level of the index. The process has five steps:

a. Risk-Free Rate

The 10-year US Treasury yield is used as the starting point, but since the US no longer holds a 'Aaa' sovereign rating, the raw yield embeds a small default premium. That premium is stripped out by netting the US CDS spread against Switzerland's CDS spread, the same benchmark for risk-free CDS spread used elsewhere in this model, to arrive at a clean dollar risk-free rate:

$$\text{Risk-Free Rate} = \text{US Treasury Yield} - (\text{CDS US} - \text{CDS Switzerland})$$

$$\text{Risk-Free Rate} = 4.61\% - (0.57\% - 0.17\%) = 4.21\%$$

b. Base Year Cash Flows

Cash flows to equity are defined as dividends plus buybacks, the total cash returned to shareholders. Rather than using the trailing 12-month payout directly, we use trailing 12-month earnings multiplied by the 10-year average payout ratio. This smooths out year-to-year volatility in buyback behaviour and produces a more stable base for projection.

$$\text{Base Year Cash Flow} = \text{Trailing 12M Earnings} \times \text{10-Year Average Payout Ratio}$$

$$\text{Base Year Cash Flow} = 234.06 \times 88.16\%$$

Table 5: Trailing 12-Month EPS (Last 4 Quarters)

Quarter	EPS
Q4 2024	\$ 58
Q1 2025	\$ 54
Q2 2025	\$ 59
Q3 2025	\$ 64

Note: Q4 2025 earnings not yet published by S&P as of May 2026.

Table 6: S&P 500 Historical Earnings & Payout Ratios (2015–2024)

Year	EPS	Payout Ratio
2015	\$ 100	107.83%
2016	\$ 106	101.66%
2017	\$ 125	88.17%
2018	\$ 153	98.51%
2019	\$ 157	93.08%
2020	\$ 140	84.90%
2021	\$ 205	80.48%
2022	\$ 199	88.74%
2023	\$ 211	78.03%
2024	\$ 232	79.91%
10-Year Average		88.16%

Source: S&P 500 reported earnings

c. Near-Term Growth Rate

Earnings are projected forward for five years using a consensus growth rate derived from EPS estimates published by major sell-side institutions. The 5-year CAGR implied by their estimates is used as the near-term growth rate.

Table 7: EPS Consensus Estimates

Institution	FY25A EPS	FY26E EPS	FY27E EPS
Goldman Sachs	\$ 272	\$ 309	342
JP Morgan	\$ 272	\$ 330	385
Bank of America	\$ 272	\$ 310	—
Deutsche Bank	\$ 272	\$ 320	—
Wells Fargo	\$ 272	\$ 310	350
Barclays	\$ 272	\$ 321	—
Implied 5Y CAGR			11.20%

Source: consensus estimates

The CAGR for five-year horizon was estimated using the consensus average growth starting from 2026 and 2027 with 16.42% and 13.42% respectively, expected to reach the risk-free rate of 4.21% by the sixth years leaving us with 6.5% for the fifth year, leaving us with an annualised CAGR of 11.20% for the next 5 years.

d. Terminal Value Growth

After year five, cash flows are assumed to grow into perpetuity at the risk-free rate used here as a proxy for the long-run nominal growth rate of the economy. This prevents the terminal value from being driven by an arbitrary assumption and ties the model's long-run growth to an observable market rate.

e. Solving for Implied Cost of Capital

With the current index level, base year cash flows, and the projected growth path in hand, the model solves for the single discount rate that makes the present value of all projected cash flows equal to the current index level of 7,403.05 as of May 1, 2026.

Cash flows are time-adjusted to reflect the point in the year at which the calculation is run, since this is computed in May, the first cash flow falls approximately 0.67 years away rather than a full year.

Table 8: S&P 500 Implied ERP — DDM Projection

	Base Year	Year 1	Year 2	Year 3	Year 4	Year 5	TV
Expected Earnings	\$ 234	\$ 260	\$ 289	\$ 322	\$ 358	\$ 398	\$ 415
Payout Ratio	88.16%	88.16%	88.16%	88.16%	88.16%	88.16%	88.16%
Cash to Equity	\$ 206	\$ 229	\$ 255	\$ 284	\$ 316	\$ 351	\$ 366
Terminal Value							\$ 9,142
Time to cashflow		0.667	1.667	2.667	3.667	4.667	4.667
Present Value		\$ 145	\$ 224	\$ 230	\$ 236	\$ 243	\$ 6,325
Intrinsic Value of Index							\$ 7,403

The resulting discount rate is the implied cost of capital for the index. Netting out the risk-free rate yields the implied ERP:

$$\text{Implied ERP} = \text{Implied Cost of Capital} - \text{Risk-Free Rate}$$

$$\text{Implied ERP} = 8.21\% - 4.21\% = 4.00\%$$

This 4.00% is the forward-looking, market-implied compensation for bearing equity risk in the US at current prices.

5. Estimating Components

a. Estimating the Relative Volatility Multiplier (Step 1)

The first thing the model needs is a multiplier that converts sovereign default spreads, which reflects the risk of default for bond-market into equity-market risk premiums. The logic is straightforward, equity investors in the same country face more volatility than bondholders, so a 1% default spread does not translate to a 1% equity premium. It needs to be scaled up by however much more volatile equities are relative to bonds.

To estimate this, daily return data is used over the period January 1, 2020 to April 2026, roughly six years spanning multiple market cycles, including the COVID-19 shock in 2020, the post-pandemic recovery in 2021, the inflation and rate-hike cycle during 2022–2023, and the subsequent stabilisation during 2024–2025.

Data construction

- **Equity proxy:** Daily adjusted closing prices of the iShares MSCI ACWI ETF, a broad market-cap-weighted index covering large and mid-cap stocks across 23 developed and 24 emerging markets.
- **Bond proxy:** Daily adjusted closing prices of the JPMorgan Funds Global Aggregate Bond Fund, a diversified portfolio of investment-grade sovereigns and corporate bonds.

Annual Volatility Construction

Annual standard deviations (σ) are computed year-by-year from the daily return series:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{\text{No. of days observed}}$$

The annual relative volatility for each year is then:

$$\lambda_{\text{year}} = \sigma_{\text{equity}} / \sigma_{\text{bond}}$$

Results:

Table 9: Annual Relative Volatility Estimates (2020–2026)

Year	σ Equity (ACWI)	σ Bond (JPM)	Relative Volatility (λ)
2020	32.14%	6.87%	4.68x
2021	12.66%	3.67%	3.44x
2022	22.78%	6.83%	3.33x
2023	12.76%	7.32%	1.74x
2024	11.95%	5.63%	2.12x
2025	17.17%	5.11%	3.36x
(Jan – Apr) 2026	9.00%	3.70%	2.43x
Average (2020–2026)	16.92%	5.59%	3.026x

The final multiplier applied in the model is $\lambda = 3.0263$, the simple average of annual relative volatilities over the full window. This value means that for every 100 basis points of default spread translates to approximately 302.63 basis points of CRP.

b. Estimating Mature Market Risk Premium (Step 2)

The US S&P 500 implied ERP is taken as an exogenous input, computed at 4.00% as of May 1st, 2026. The full derivation is documented in the preceding section.

Since the US no longer holds a 'Aaa' sovereign rating, its implied ERP already reflects a small country risk premium of its own. To extract a pure mature-market baseline, one that represents a truly risk-free equity market, the US CRP is backed out:

$$ERP (mature) = ERP (US implied) - CRP (US CDS \times Relative Volatility)$$

$$ERP (mature) = 4.00\% - (0.33\% \times 3.026x) = 3.00\%$$

This adjusted ERP is the base to which all country-specific CRPs are added, ensuring that countries with zero sovereign risk arrive at exactly the mature-market ERP.

c. Estimating Default Spread (Step 3)

The model estimates sovereign default spreads using three methods, each designed for a different data environment.

Method A: CDS-Based Default Spread

Where 10-year CDS markets exist, they are the first choice. CDS spreads are market-determined and forward-looking. They reprice in real time as sovereign risk shifts, which makes them theoretically superior to ratings that can lag events by months. Switzerland's CDS spread is netted out to isolate the country-specific component:

$$Default Spread (CDS) = CDS_{country} - CDS_{Switzerland}$$

Switzerland is chosen as the risk-free sovereign benchmark as we assume it as the world's safest sovereign with trading CDS spread, reflecting near-zero default probability, political neutrality, and deep capital markets.

Table 10: Credit Default Swaps of Selected Countries

Country	CDS	CDS Net of Switzerland
Switzerland	0.17%	0.00%
United States	0.50%	0.33%
India	0.84%	0.67%

Source: Damodaran Online (NYU Stern School of Business), Last Updated November 2025

Method B: Rating-Based Default Spread

For countries where a Moody's sovereign exists but no CDS data, a lookup table maps each letter grade to an annualised default spread in basis points. The table spans from Aaa at 0 bps to C at 1,750 bps, is sourced from Damodaran's annual sovereign rating update. The rating-to-spread mapping reflects the historical average CDS of countries in each rating category.

Table 11: Moody's Rating-to-Default-Spread Mapping

Moody's	Default Spread
Aaa	0
Aa1	40
Aa2	49
Aa3	59
A1	70
A2	84
A3	119
Baa1	158
Baa2	189
Baa3	218
Ba1	248
Ba2	298
Ba3	356
B1	446
B2	545
B3	644
Caa1	743
Caa2	892
Caa3	991
Ca	1188
C	1750
NR	NA

Source: Damodaran Online (NYU Stern School of Business)

Method C: PRS Based Default Spread

For countries that are neither rated by Moody's and lack a CDS market, typically lower-income frontier nations, the Political Risk Service (PRS) composite score is used to infer a default spread. The PRS uses twenty-two political, financial, and economic risk variables to generate a composite country risk score ranging from 0 to 100, where lower scores indicate higher country risk and higher scores indicate lower country risk.

For these countries, a lookup table maps ranges of PRS scores to synthetic sovereign ratings and corresponding default spreads in basis points. The mapping spans from PRS score ranges of 85–90, corresponding to a 0-basis point default spread, to score ranges of 0–50, corresponding to a 1,750-basis point default spread.

The score-to-spread relationship is computed using the observed sovereign ratings and default spreads of rated countries exhibiting similar PRS scores, thereby allowing the estimation of synthetic sovereign default spreads for unrated frontier markets. This framework is sourced from Professor Aswath Damodaran's country risk premium methodology.

Table 12: PRS Scores-to- Default Spread Mapping

More than	Less than	Synthetic Rating	Default Spread
0.000	50.000	Below Ca	17.50%
50.001	55.000	Ca	11.88%
55.001	57.000	Caa3	9.91%
57.001	60.000	Caa2	8.92%
60.001	62.000	Caa1	7.43%
62.001	64.000	B3	6.44%
64.001	66.000	B2	5.45%
66.001	68.000	B1	4.46%
68.001	69.000	Ba3	3.56%
69.001	72.000	Ba2	2.98%
72.001	74.000	Baa2	2.48%
74.001	76.000	Baa1	2.18%
76.001	80.000	A2	1.89%
80.001	82.500	Aa3	0.59%
82.501	85.000	Aa1	0.40%
85.001	90.000	Aaa	0.00%

Source: Damodaran Online (NYU Stern School of Business)

d. Estimating Country Risk Premium (Step 4)

For each country, the CRP is computed by multiplying the relevant default spread by the relative volatility multiplier $\lambda = 3.0263$. Formally:

$$CRP = \text{Default Spread} \times \lambda$$

The multiplication by λ does the conversion, as it scales up the bond-market default spread into an equity-market risk premium, reflecting that equity holders bear proportionally greater country risk than bondholders for the same sovereign.

A worked example for India, illustrates how the three methods diverge.

India's CDS net of Switzerland = 0.67%, Moody's Rating = Baa3 & PRS Score (As of Last Updated at September 2021) = 72

Table 13: CRP Estimation for Each Method

CRP Variant	Default Spread	RV Multiplier	CRP
Method A (CDS Net of Switzerland)	0.6700%	3.0263x	2.0276%
Method B (Rating-Based)	2.1768%	3.0263x	6.5876%
Method C (PRS Score-Based)	2.4794%	3.0263x	7.5035%

e. Estimating Equity Risk Premium (Step 5)

For each country, the ERP is computed by adding the mature market risk premium of 3.00% computed earlier. Formally:

$$ERP = \text{Mature Market Risk Premium} + CRP$$

As a worked example, for India under each method,

Table 14: ERP Estimation for Each Method

ERP Variant	Mature Market ERP	CRP	ERP
Method A	3.0013%	2.0276%	5.0290%
Method B	3.0013%	6.5876%	9.5889%
Method C	3.0013%	7.5035%	10.5048%

f. Data Hierarchy & Final Estimate Selection

Since all three methods produce different estimates, the model needs a rule for which one to use. The hierarchy is straightforward, the most market-responsive source takes priority, with each subsequent method serving as a fallback when data is unavailable.

Table 15: Data Hierarchy & Estimation Priority

Priority	Method	Rationale
1 st	Method A	Market Implies, & Forward Looking
2 nd	Method B	Fallback on Average CDS Spread of Sovereign with same Rating
3 rd	Method C	Fallback on Country Risk Scores when Neither Rated nor CDS Traded

Where CDS data exists, it is used, as it prices in real time and carries no lag. Where it does not, ratings provide a reasonable proxy. And for the frontier markets that fall through both nets, PRS scores ensure the model still produces an estimate rather than a blank.

It is worth being direct about what this hierarchy implies for interpretation. The three methods do not converge, and the India example makes that visible. Method A gives India a CRP of 2.03%, while Method C gives 7.50%, a gap of over 550 basis points. The divergence is not a model flaw; it reflects the limitations of each fallback method relative to the gold standard.

CDS spreads are the most theoretically sound input: they are market-determined, forward-looking, and function as a direct insurance price against sovereign default. Method B uses the average CDS spread of all countries in the same rating class, which smooths out country-specific signals and introduces rating-agency lag. Method C is the weakest of the three, PRS scores are judgment-based, vary across services, and produce results that are at times difficult to reconcile with observable reality. Where CDS data exists, it should be used and trusted. Methods B and C exist purely to extend coverage to countries where the market has not spoken, not to compete with it.

6. Regional CRP & ERP

Country-level CRP & ERP estimates are aggregated to regional averages using GDP-weighted averaging. This approach assigns greater weight to larger economies within each region, reflecting the capital allocation pattern of a globally diversified investor. Formally:

$$CRP (region) = \Sigma [CRP i \times GDP i] / \Sigma GDP i$$

Equivalently,

$$ERP (region) = \Sigma [ERP i \times GDP i] / \Sigma GDP i$$

Table 16: GDP-Weighted Regional ERP Estimates, April 2026

Region	GDP (USD mn)	Regional CRP	Regional ERP
North America	3,09,94,593	0.97%	3.97%
Western Europe	2,35,66,520	1.47%	4.47%
Australia & New Zealand	20,17,879	0.05%	3.05%
Asia	3,48,57,584	2.28%	5.28%
Middle East	40,99,199	6.08%	9.08%
Eastern Europe & Russia	56,65,313	7.95%	10.95%
Central & South America	65,63,150	9.77%	12.77%
Africa	27,94,308	16.47%	19.47%
Caribbean	3,56,924	19.98%	22.98%
World (GDP-Weighted)	11,09,15,469	2.99%	5.99%

The world aggregate CRP of 2.99% reflects the GDP-weighted blend across all covered economies. The number is anchored down by the large developed-market weight of North America, Western Europe, and Asia, which together account for the bulk of global GDP, but the regional spread tells a starker story, an investor moving from Australia & New Zealand at 0.05% CRP to the Caribbean at 19.98% CRP is crossing nearly 2,000 basis points of additional required return. That gap is the practical case for why country risk premiums matter.

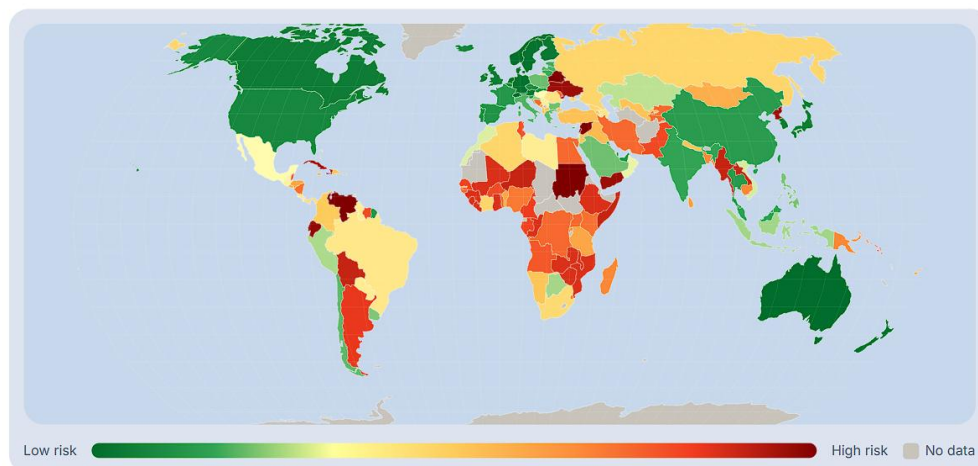
a. Estimated Country Risk Premiums

The geographic pattern is not a coincidence. Compare the map below with the three in the Country Risks section, the congruence is hard to miss. Countries the Democracy Index flags as authoritarian, the Peace Index marks as conflict-prone, and Transparency International scores as deeply corrupt tend to have higher CRPs here.

The one divergence worth noting is China, here, milder than what the Democracy and Corruption indices would suggest. That is not a model error but CDS compression at

work. China's investment-grade sovereign status and strong foreign capital inflows keep its CDS spread tighter than its structural risk profile would imply.

Figure 4: Estimated Country Risk Premiums Heatmap



7. Conclusion

The problem this paper started with is straightforward: investors are risk averse, and the return they demand depends on where and what they are investing in. A mature-market equity risk premium captures that demand for a Danish or Australian investor buying into deep, transparent, well-governed markets. It does not capture it for an analyst pricing a company operating in Nigeria, or building a DCF for an acquisition in Venezuela. The risks are simply different: political instability, exposure to violence, corruption, and weak legal systems in emerging markets create layers of uncertainty that mature markets have largely priced away. A discount rate that ignores those differences is not conservative, it is mispricing.

The historical approach to estimating that premium only compounds the problem. As illustrated using the Nifty 50 data in table 1, the choice of window and averaging method alone can swing an estimate by hundreds of basis points, and even if both choices were made perfectly, the output is still backward-looking. The implied approach fixes that, but only where the inputs exist. For most of the world especially in emerging markets, the input does not exist. If they do, the results are probably going to be misleading as the valuations in these economies are mainly driven by growth narratives and carried by growth expectation, not because the fundamentals are strong.

While the framework documented here does not resolve all of that complexity; no model can. But it does make the choices explicit. The mature market baseline is derived from observable market data, the country risk premiums are scaled to equity risk, not bond market default spreads. A transparent estimation hierarchy: CDS where markets exist, ratings where they do not, and country risk scores as the last resort for markets that have never been formally assessed. Every assumption is explicit, every departure from the standard framework is explained. The outputs are directly usable in any cross-

border valuation without rebuilding the methodology from scratch, whether they should be utilised is a judgment call that depends on the specific valuation context, but that judgment is now easier to make, because the methodology behind the numbers is no longer a black box.

8. Key Assumptions

a. Single Volatility Multiplier

The model applies a uniform relative volatility multiplier of 3.0263x across all 174 countries. In reality, equity-bond volatility ratios differ across markets, but estimating country-specific multipliers for frontier markets with thin or non-existent equity markets is not feasible. A global ACWI/bond ratio is consistent with the perspective of a globally diversified investor.

b. Switzerland as the Risk-Free CDS Benchmark

Netting CDS spreads against Switzerland rather than zero removes the common global credit risk component embedded in all CDS spreads, producing a cleaner measure of country-specific default risk.

c. US Implied ERP as the Mature Market Anchor

The US implied ERP, adjusted for its own CRP, at 3.00% is used as the mature market baseline; the floor to which all country-specific CRPs are added.

d. GDP Weighting for Regional Aggregation

GDP weighting is preferred over equal weighting because it reflects economic significance and prevents small, high-risk economies from disproportionately inflating regional averages.

9. Limitations

a. Data Staleness

The three data sources are not contemporaneous. CDS spreads are from November 2025, Moody's ratings from January 2026, and PRS scores from September 2021. The PRS gap is the most significant, for frontier markets where PRS is the only available signal, the model is pricing 2026 risk with 2021 data. Users should be aware that CRP estimates for these countries are more sensitive to data vintage than those derived from CDS or ratings.

b. CDS Compression in Investment-Grade Emerging Markets

Method A is the most theoretically sound but not immune to distortion. In benign market conditions, CDS spreads for investment-grade emerging economies are compressed by foreign capital inflows and global liquidity conditions, factors unrelated

to the underlying structural risk of operating a business there. The China example in figure 4 illustrates this directly. Users valuing long-duration assets in such markets should consider whether Method A alone understates the true risk premium.

c. Single Volatility Multiplier

Applying a uniform λ of 3.0263x across all markets is a necessary simplification, but it means the model does not capture the fact that equity markets in frontier economies are structurally more volatile relative to their bond markets than the global average implies. The multiplier may understate CRP for the most illiquid markets and overstate CRP for mature developed markets.

d. PRS Subjectivity

PRS scores are judgment-based and can produce counterintuitive results. They exist in this model purely to extend coverage to unrated frontier markets, not as a preferred signal. Estimates derived from Method C carry wider uncertainty than those from Methods A or B.

10. Data Source

Table 17: Data Sources

Data Item	Source	Date / Period
US Implied ERP (S&P 500)	User input (model)	May 2026
Moody's Sovereign Ratings	Moody's Credit Ratings	January 2026
Rating-to-Default-Spread Table	NYU Stern / Damodaran Online	January 2026
10-Year CDS Spreads	NYU Stern / Damodaran Online	November 2025
PRS Scores	NYU Stern / Damodaran Online	September 2021
ACWI Equity Returns	iShares MSCI ACWI ETF (ACWI)	Jan 2020 – Apr 2026
Global Bond Returns	JPMorgan Global Aggregate Bond Fund	Jan 2020 – Apr 2026
Country GDP	World Bank	2024

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